

ENTERPRISE DATA GOVERNANCE FRAMEWORK IMPLEMENTATION FOR E-COMMERCE PLATFORM

DATA GOVERNANCE TARGET OPERATING MODEL (TOM)

1. EXECUTIVE SUMMARY

The Data Governance Target Operating Model (TOM) defines the structure, roles, processes, and technologies required to manage data as a strategic asset across the enterprise. It ensures alignment between business objectives, regulatory requirements, and data management practices.

2. OBJECTIVES

- Establish clear accountability for data ownership and stewardship
- Improve data quality and trust across all business domains
- Enable regulatory compliance and auditability
- Standardize data governance processes across the enterprise
- Support scalable governance aligned with modern data platforms

3. OPERATING MODEL OVERVIEW

3.1. Governance Model Type

✓ Federated Governance Model

- Central governance defines policies & standards
- Domain teams execute governance within business units

4. GOVERNANCE STRUCTURE

4.1. Organizational Layers

Executive Layer

- Chief Data Officer (CDO)
- Data Governance Council

Strategic Layer

- Data Governance Lead
- Domain Data Owners

Operational Layer

- Data Stewards
- Data Custodians (IT)

Supporting Functions

- Risk & Compliance
- Legal
- Security

4.2. Roles & Responsibilities

Role	Responsibility
Data Owner	Accountable for data domain quality & compliance
Data Steward	Ensures data quality, metadata, definitions
Data Custodian	Technical management of data
Governance Council	Decision-making & policy approval

5. RACI FRAMEWORK

Activity	Owner	Steward	Custodian	Governance Council
Data Definition	A	R	C	I
Data Quality Monitoring	A	R	C	I
Access Control	A	C	R	I
Policy Approval	R	C	I	A
Metadata Management	A	R	C	I
Issue Resolution	A	R	C	I

(R = Responsible, A = Accountable, C = Consulted, I = Informed)

6. CORE GOVERNANCE PROCESSES

6.1. Data Quality Management

- Define critical data elements (CDE)

Determination criteria:

- Regulatory/reporting impact
- Revenue/payment impact
- Customer experience impact
- Fraud/risk impact
- Operational dependency across domains

Example of e-commerce CDE: customer_id, email, order_id, payment_status, sku, vendor_id.

- Establish data quality rules & thresholds
- Monitor via dashboards
- Issue management workflow:
 - Detect → Log → Assign → Resolve → Validate → Close

6.2. Metadata Management

- Maintain enterprise data catalog
- Define business glossary
- Track data lineage (source → transformation → consumption)

6.3. Data Access & Security

- Role-Based Access Control (RBAC)
- Attribute-Based Access Control (ABAC) (optional advanced)
- Access request & approval workflow
- Periodic access review

6.4. Data Lifecycle Management

- Define lifecycle stages:
 - Create → Store → Use → Archive → Delete
- Retention policies based on regulatory requirements

6.5. Data Classification

- Define classification levels:
 - Public
 - Internal
 - Confidential
 - Restricted (PII / highly sensitive)

7. GOVERNANCE ARTEFACTS

Mandatory Artefacts:

- Data Governance Policy
- Data Standards
- Data Quality Rules
- Data Classification Policy
- Data Access Policy
- Data Dictionary / Glossary
- Data Lineage Documentation
- Governance Control Register

8. TECHNOLOGY & TOOLING

8.1. Data Governance Tools

- Data Catalog: Collibra / Alation / Atlan / Microsoft Purview
- Data Quality: Informatica / Great Expectations

8.2. Data Platforms

- Data Warehouse (Snowflake, BigQuery)
- Data Lake / Lakehouse (Databricks)

8.3. Integration

- Metadata ingestion from ETL pipelines
- Automated lineage tracking

9. INTEGRATION WITH DATA ARCHITECTURE

- Governance embedded in:
 - Data ingestion pipelines
 - Transformation layers
 - BI / reporting layer
- Controls applied at:
 - Data creation
 - Data storage
 - Data access

10. COMPLIANCE & RISK MANAGEMENT

- Align with regulatory frameworks (e.g., GDPR principles)
- Implement audit trails
- Periodic compliance assessment
- Risk identification and mitigation

11. KPIs & SUCCESS METRICS

Metric	Description
Data Quality Score	>95% of data meeting quality standards
Issue Resolution SLA	>95% of issues are resolved within the SLA according to severity
Data Availability	>99.5% availability for Tier-1 datasets during business hours / agreed SLA
Metadata Completeness	>98% of critical datasets documented in the catalog
Access Review Completion	100% quarterly completion for privileged roles

KPI Definitions:

KPI	Formula (Example)	Target Phase 1–2	Target Steady State
DQ Score	% CDE records that pass all quality rules	≥90%	≥95%
Metadata Completeness	% critical data assets with owner, steward, definition, classification, lineage	≥80%	≥98%
Owner Coverage	% critical data assets with Data Owner	≥95%	100%
Issue Resolution SLA	% of issues are resolved within the SLA according to severity	≥80%	≥95%
Access Review Completion	% quarterly access reviews completed on time	≥90%	100%

12. IMPLEMENTATION ROADMAP**Phase 1: Assessment**

- Current state analysis
- Gap identification

Phase 2: Design

- Define TOM
- Develop policies & standards

Phase 3: Pilot

- Implement in selected domain

Phase 4: Rollout

- Scale across enterprise

Phase 5: Optimization

- Continuous improvement

13. CHANGE MANAGEMENT & ADOPTION

- Stakeholder engagement plan
- Training & awareness programs
- Governance champions per domain
- Communication strategy

14. ESCALATION FRAMEWORK

- Data issues escalated:
 - Steward → Owner → Governance Council
- SLA for resolution defined

SLA Table (Example):

Severity	Initial Response	Target Resolution
Critical	< 4 jam	< 24 jam
High	< 1 hari kerja	< 5 hari kerja
Medium	< 2 hari kerja	< 10 hari kerja
Low	< 5 hari kerja	< 30 hari kerja

15. REVIEW & CONTINUOUS IMPROVEMENT

- Quarterly governance review
- Annual TOM update
- Continuous KPI monitoring

16. APPENDICES

Appendix A: Data Quality Dimensions

- Accuracy
- Completeness
- Consistency
- Timeliness

Appendix B: Domain-Centric Governance

Domain	Data Owner	Steward	Custodian
Customer	Head of CRM/Marketing	CRM Steward	Data Platform/IT
Product	Merchandising Lead	Catalog Steward	IT
Order & Transaction	Operations/Payments Lead	Order Steward	IT
Vendor	Supply Chain Lead	Vendor Steward	IT

Appendix C: Governance Cadence

Cadence	Forum	Purpose
Monthly	Domain Governance Review	DQ trends, open issues, access exceptions
Quarterly	Governance Council	Policy decisions, cross-domain conflicts, KPI review
Annually	Policy/TOM Review	Regulatory changes, roadmap reprioritization

Appendix B: Sample Workflow

- Data Issue Management Workflow Diagram

Visual Diagram (Architecture + Governance Flow)

